Example: (Union)

E = 0

F = 1

E+F = 0+1 regular expression

L(E) = {0}

L(F) = {1}

L(E) ∪ L(F) = {0} ∪ {1} = {0,1}

L(E+F) = L(0+1) = {0,1}

∴ L(E+F) = L(E) ∪ L(F)

Example: (Concatenation)

E = 0

F = 1

EF = 01 regular expression

L(E) = {0}

L(F) = {1}

L(E) L(F) = {0}{1} = {01}

L(EF) = L(01) = {01}

∴ L(EF) = L(E) L(F)

Example: (closure)

E = 1

L(E) = {1}

$L(E^*) = L(1^*) = \{\epsilon, 1, 11, 111, \ldots\}$

$(L(E))^* = \{1\}^* = \{\epsilon, 1, 11, 111, \ldots\}$

$\therefore L(E^*) = (L(E))^*$

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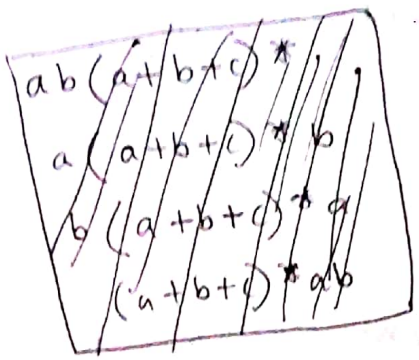
Write regular expression for the following language
All strings both begin and end with 0 over the alphabet {0, 1}

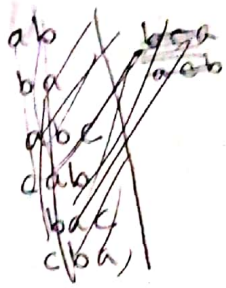
Ans:

$$0(10)^*$$

Write regular expressions for the following languages
The set of strings over alphabet {a, b, c} containing at least one a and at least one b.

Ans:





~~ab+(a+b+c)*+ba~~

$(ab+ba)(a+b+c)^*$

Find L^*

$L = \{0, 11\}$

$L^0 = \{\epsilon\}$

$L^1 = \{0, 11\}$

$L^2 = \{00, 011, 110, 1111\}$

$L^3 = \{000, 0011, 0110, 1100, 01111, 11011, 11110, 111111\}$

⋮

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